MarketSage AI

Master Engineering Blueprint — **Part III**

Feature Store and Prediction Models

Feature catalogue across six groups, train/serve parity, triple-barrier labelling, purged cross-validation, the model portfolio, ensembling, calibration and retraining policy.

This is an extract of the MarketSage AI master plan, published separately for convenience. Chapter, section and equation numbering matches the full document, and cross-references point back into it. Read Chapter 3 (Reality Calibration) first — every design decision in this part follows from it.

CHAPTER 8

Feature Store and Feature Catalogue

8.1 Why a feature store and not a script

The most common cause of a backtest that works and a live system that does not is **train/serve skew**: the offline feature was computed differently from the online one. The rule is absolute.

One definition, two materialisations

A feature is defined *once*, in one function, with an explicit dependency list and an as-of timestamp. The offline path runs it over Parquet history; the online path runs the same function over Redis state. A CI test replays a live day offline and asserts every feature matches to within floating-point tolerance. If it does not match, the build fails.

Listing 8.1: Feature definition pattern

```python
@feature(name="ofi_60s", version=3, deps=["book.bid","book.ask"],
        window="60s", ttl="5s", fill="ffill_max_2")
def order_flow_imbalance(book: BookWindow) -> float:
    """Signed order-flow imbalance over the trailing 60 seconds.
    Point-in-time safe: reads only data with ts <= as_of."""
    ...

# Offline: engine.materialise(ofi_60s, start, end) -> Parquet column
# Online : engine.serve(ofi_60s, as_of=now) -> float from Redis
# CI : assert_offline_online_parity(ofi_60s, sample_day)
```

8.2 Feature catalogue

8.2.1 Group A — Price and trend (the baseline)

Feature	Windows	Rationale
Log returns	1, 2, 5, 15, 30, 60 min	Raw momentum at several scales.
Return z-score vs time-of-day baseline	same	Normalises the U-shaped intraday volatility profile. **Critical.**
Realised volatility (Parkinson, Garman–Klass, Rogers–Satchell)	5, 15, 30, 60 min	Range-based estimators are far more efficient than close-to-close.
Bipower variation and jump component	30 min	Separates continuous volatility from jumps; events cause jumps.
Distance from VWAP, from opening range, from previous close	session	Standard intraday anchors that real traders actually watch.
Position within the day's range	session	Mean-reversion versus breakout context.
Trend strength / efficiency ratio	15, 30 min	Distinguishes a directional move from chop; regime information.
Autocorrelation of 1-min returns	30 min	Negative values signal a mean-reverting microstructure regime.

8.2.2 Group B — Microstructure (where the short-horizon edge lives)

Feature	Windows	Rationale
Order-flow imbalance (OFI)	10, 30, 60 s	Best-in-class short-horizon directional predictor when book data is available.
Bid–ask spread and its z-score	30 s	Widening spreads precede volatility and raise transaction cost — feeds both the model and the cost model.
Depth imbalance at top levels	instantaneous	Directional pressure.
Trade-sign imbalance (tick rule / Lee–Ready)	60 s	Aggressor direction.
Volume-clock and VPIN-style toxicity proxy	50 buckets	Flow toxicity spikes before large moves.
Large-trade ratio	60 s	Institutional participation proxy.
Quote update intensity	10 s	Attention and uncertainty proxy.
Futures basis (future minus spot)	instantaneous	Basis moves often lead spot.

Availability caveat

Full order-book depth may not be available on a retail broker feed. Design the model so Group B is optional: train two variants, with and without microstructure, and let the serving layer select by feed capability. Never train on features you will not have at 09:16 on a live Tuesday.

8.2.3 Group C — Options-derived (the most differentiated group)

Feature	Update	Rationale
ATM implied volatility, both expiries	1–5 s	Level of expected volatility.
IV term structure slope (near vs next expiry)	5 s	Inversion signals expected near-term event risk.
25-delta risk reversal (skew)	5 s	The market's directional fear; one of the strongest single option-derived predictors.
Butterfly / smile convexity	5 s	Tail pricing.
IV rank and IV percentile	daily	Is volatility cheap or expensive in context?
Put-call ratio, by volume and by open interest	1 min	Positioning and sentiment.
Open-interest change by strike, call and put	1 min	Where new positions are being built — writers define support and resistance.
Max pain and distance to it	5 min	Expiry-week pinning effect.
Estimated dealer gamma exposure (GEX)	1 min	Positive gamma dampens moves, negative gamma amplifies them. **Regime-defining.**

Implied move for the remaining session	1 min	The denominator of the trade filter in Chapter 3.
Volatility risk premium (IV minus forecast realised)	5 min	The core edge for volatility structures.

8.2.4 Group D — Cross-asset and global

Overnight and same-session values of: global equity futures, regional indices, crude, gold, dollar index, USD/INR, US yields, and a global-risk composite (first principal component of a basket of risk assets). Include both level changes and z-scores versus their own trailing distributions.

8.2.5 Group E — Event features

Feature	Note
Event class one-hot / embedding	Learned embedding is better once enough history exists.
Calibrated severity, polarity, novelty, credibility, confidence	From Chapter 6.
Surprise z-score	For scheduled releases; the single most powerful macro feature.
Prior impact posterior (mean, spread, n_{eff})	From Chapter 7. Feeding the prior in as a feature lets the model learn when to trust it.
Seconds since event	The decay clock.
Event density in trailing 30 min	Multiple simultaneous events change dynamics.
Conflict flag	Opposing events present.

8.2.6 Group F — Calendar, regime and session

Minutes since open, minutes to close, day of week, days to weekly expiry, days to monthly expiry, expiry-day flag, budget/policy-day flag, holiday-adjacent flag, current regime label from the regime model, trailing 20-day realised volatility percentile, and recent trend regime.

Feature hygiene

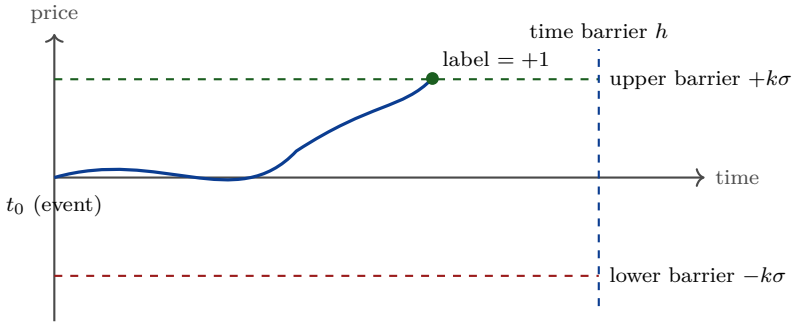
Target roughly 120–250 engineered features, not thousands. Run **Purged Group Time-Series CV** feature importance and drop anything that does not survive. High-cardinality raw features and features with any dependence on future information are removed automatically by a leakage linter that checks each feature's dependency timestamps.

CHAPTER 9

Prediction Models

9.1 Labelling: the decision that determines everything

Predicting "next candle up or down" is the wrong label because it does not correspond to a tradeable decision. Use the **triple-barrier** method.



Choice	Specification
Barriers	Upper and lower at $\pm k\sigma_t$ where σ_t is the *current* volatility estimate (not a fixed point value). k tuned per horizon; start with $k = 1.5$.
Time barrier	$h \in \{5, 15, 30\}$ minutes and always before the session close.
Label	$+1$ upper hit first, -1 lower hit first, 0 time barrier hit.
Sample weights	Weight each sample by uniqueness (overlapping outcome windows are not independent) and by absolute realised return.
Meta-labelling	A second model predicts *whether the primary model's call should be taken*. Primary model decides side; meta model decides size, including size zero. This consistently improves precision.

Why triple-barrier changes the product

It directly encodes stop-loss and target into the label. A model trained this way is answering the question you actually care about — "if I enter now with this stop and this target, do I win?" — rather than an academic question about candle colour.

9.2 Cross-validation without lying to yourself

Technique	Why it is mandatory
Purged K-fold	Remove training samples whose outcome window overlaps the test window. Without purging, the model sees the answer.
Embargo	Additionally drop a buffer (e.g. 1 trading day) after each test fold to kill serial-correlation leakage.
Walk-forward (anchored and rolling)	The only evaluation that resembles production. Report both.
Combinatorial purged CV	Generates many backtest paths, giving a *distribution* of Sharpe ratios rather than one lucky number.

Deflated Sharpe ratio	Corrects for the number of strategy variants tried. If you tested 200 configurations, your best Sharpe is inflated — quantify it.
Event-level splitting	Never split the same event's samples across train and test. Group by event ID.

Leakage checklist — run before believing any result

1. Does any feature use data with a timestamp after the decision time?
2. Is normalisation (mean/std, scaler) fitted on the full dataset instead of train only?
3. Are volatility estimates computed with a centred window?
4. Are labels from overlapping windows present in both train and test?
5. Was feature selection done before the split?
6. Does the news timestamp reflect publication or the time you downloaded it?
7. Is the option chain snapshot the one available at decision time, or the end-of-day file?
8. Are corporate actions and index reconstitutions handled point-in-time?

Any single "yes" invalidates the result. Most spectacular backtests die here.

9.3 Model portfolio

Do not build one model. Build a small ensemble of models with genuinely different inductive biases, then combine them.

Model	Predicts	Role and notes
M0 Rules baseline	direction	Simple event-prior lookup plus momentum filter. **Every ML model must beat this or it does not ship.** Also the degraded-mode fallback.
M1 LightGBM	$P(\text{up})$, triple-barrier class	The workhorse. Handles tabular heterogeneous features and missing values, trains in minutes, is interpretable via SHAP. Expect this to be your best model for a long time.
M2 Quantile LightGBM	$q_{05}, q_{25}, q_{50}, q_{75}, q_{95}$ of r_h	Pinball loss. Feeds the fan chart directly.
M3 Temporal CNN / TCN	direction and magnitude	Learns local shape patterns from raw candle and microstructure sequences that handcrafted features miss.
M4 Sequence transformer	multi-horizon distribution	Attention over the last N minutes plus event tokens. Highest ceiling, highest overfitting risk, needs the most data. **Build this last, not first.**
M5 Volatility model	σ_{t+h}	HAR-RV baseline plus a gradient-boosting correction on the residual. Volatility is the most forecastable target — give it a dedicated, well-tuned model.

M6 model	IV-change	$\Delta\sigma_{IV}$	Predicts implied volatility movement. **Directly determines options profitability.** Features: current IV rank, event class, term-structure slope, time to expiry, realised-implied spread.
M7	Meta-labeller	take / skip	Trained on primary model outputs plus market state. Its job is to say no.
M8	Regime classifier	regime label	Hidden Markov model or clustering over volatility, trend and liquidity. Everything else is conditioned on it.

9.4 Ensembling and calibration

1. **Stacking with time-aware folds.** Out-of-fold predictions from M1–M4 become inputs to a simple logistic regression meta-learner. Keep the meta-learner simple; a complex stacker overfits immediately.
2. **Regime-conditional weights.** Model weights vary by regime from M8; a transformer that shines in trending regimes may be useless in chop.
3. **Probability calibration.** Fit isotonic regression or Platt scaling on a held-out, chronologically later slice. Then verify with a reliability diagram and the Brier score decomposition (reliability, resolution, uncertainty).
4. **Conformal prediction.** Wrap the magnitude forecast in split conformal intervals to obtain finite-sample coverage guarantees. When the interval is too wide to be tradeable, the honest answer is no trade — and now the system can prove it.

The calibration contract

The dashboard's headline metric is not accuracy. It is the reliability diagram: of all the times the system said 60%, how often was it right? A system that is calibrated can be sized correctly. A system that is accurate but overconfident will still blow up.

9.5 Training loop and retraining policy

Job	Cadence	Detail
Nightly feature materialisation	Daily 18:00	Recompute the day's features, run quality gates, append to Parquet.
Incremental retrain	Weekly (weekend)	Refit M1, M2, M5, M6 on the expanded window; run walk-forward; promote only if it beats the incumbent on the last 4 weeks out-of-sample.
Full retrain and architecture search	Quarterly	Includes M3/M4 hyperparameters and feature re-selection.
Impact library rebuild	Weekly	New events added; posteriors updated.
Calibration refresh	Weekly	Isotonic refit on the most recent 4 weeks.
Drift monitor	Continuous	PSI on features, KL divergence on prediction distribution, rolling Brier score. Breach triggers alert and blocks auto-execution.

Champion/challenger	Continuous	The challenger runs in shadow mode on live data for at least 15 sessions before it may be promoted.

Anti-overfitting discipline

Keep a **locked holdout**: the most recent 3 months of data, encrypted and untouched. You may evaluate against it *once*, at the Phase 7 gate. Every additional peek burns the only honest estimate you will ever have. Log every experiment; deflate your Sharpe by the number of trials.